

EE3621: Digital Signal Processing

Lecture 1: Course Introduction & Discrete-Time Signals (III B.Tech EEE)

Lecture Notes (40-Minute Plan)

1 Lecture Plan & Objectives (40 Minutes Breakdown)

- **00:00 – 05:00 (5 mins):** Welcome, Course Objectives, & Syllabus Walkthrough
 - **05:00 – 18:00 (13 mins):** Introduction to DSP, Analog vs. Digital Processing (Block Diagram, Pros & Cons)
 - **18:00 – 30:00 (12 mins):** Classification of Discrete-Time (DT) Signals (Periodicity, Energy/Power, Symmetry)
 - **30:00 – 38:00 (8 mins):** Elementary Sequences (Impulse, Step, Ramp, Sinusoid, Exponential)
 - **38:00 – 40:00 (2 mins):** Q&A and Summary
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2 Introduction to DSP & Analog vs. Digital Processing

A **signal** is any physical quantity that varies with time, space, or any other independent variable, carrying information. In electrical engineering, examples include voltage waveforms $v(t)$, alternating current $i(t)$, and power grid frequency.

- **Analog Signal Processing (ASP):** Signals are processed in their continuous-time, continuous-amplitude form using analog hardware such as resistors (R), inductors (L), capacitors (C), and operational amplifiers.
- **Digital Signal Processing (DSP):** Signals are represented as sequences of numbers (discrete-time, quantized amplitude) and processed using digital processors (microprocessors, FPGAs, or DSP chips).

2.1 Typical DSP System Block Diagram

An analog input signal $x(t)$ is first converted to a digital sequence, processed mathematically, and then optionally converted back to analog:



Figure 1: Block diagram of a typical real-time DSP system.

2.2 ASP vs. DSP Comparison

Feature	Analog Signal Processing (ASP)	Digital Signal Processing (DSP)
Components	Resistors, inductors, capacitors, Op-Amps.	Multipliers, adders, registers, software.
Accuracy	Limited by component tolerances (e.g., $\pm 5\%$) and drift.	Highly accurate; limited only by word-length.
Flexibility	Hard to modify; requires physical design changes.	Extremely flexible; simple software updates.
Storage	Hard to store without degradation.	Easy storage on flash/digital media.
Speed	Works at very high frequencies (real-time).	Limited by sampling rate and clock speed.

Table 1: Comparison of ASP and DSP.

2.3 Analog Pre-Processing: AAF and S&H Circuits

Before an analog signal can be sampled and digitized, it must pass through two physical analog circuits:

1. **Anti-Aliasing Filter (AAF):** This is typically a passive or active RC low-pass filter. It attenuates frequency components above the Nyquist frequency $f_N = f_s/2$. The RC cutoff frequency f_c must satisfy:

$$f_c = \frac{1}{2\pi RC} \leq \frac{f_s}{2} \implies RC \geq \frac{1}{\pi f_s} \quad (1)$$

If this condition is violated, out-of-band noise folds back into the passband, causing aliasing.

2. **Sample-and-Hold (S&H) Circuit:** This circuit captures the continuous analog voltage and holds it steady for quantization. It consists of a MOSFET switch with on-resistance R_{on} , a hold capacitor C_{hold} , and an op-amp buffer.

- **Tracking Mode:** Switch is closed. The capacitor charges with time constant $\tau_{acq} = R_{on}C_{hold}$. To achieve 0.1% accuracy (for a 10-bit ADC), the tracking time T_{track} must be:

$$T_{track} \geq 7\tau_{acq} = 7R_{on}C_{hold} \quad (2)$$

- **Hold Mode:** Switch is open. The capacitor discharges through the buffer input resistance R_{in} with time constant $\tau_{hold} = R_{in}C_{hold}$. To prevent signal voltage decay (droop), we require:

$$R_{in}C_{hold} \gg T_s = \frac{1}{f_s} \quad (3)$$

2.4 Visualizing Signal Types

To understand the transformation of a signal through a DSP system, we can observe the difference between an analog signal, a sampled (discrete-time) signal, and a digital (quantized) signal.

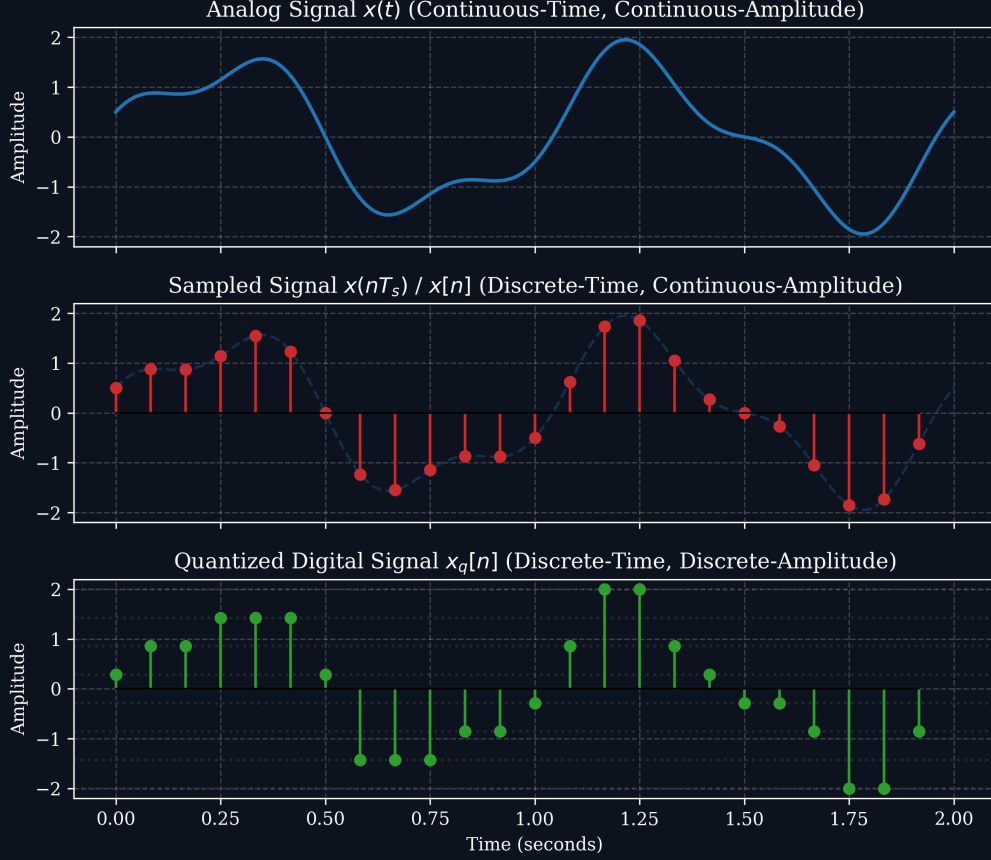


Figure 2: Analog signal $x(t)$, sampled sequence $x[n]$, and quantized digital sequence $x_q[n]$.

2.5 Quantization Resolution and Bit Depth

Quantization is the process of converting a continuous-amplitude signal into a discrete-amplitude signal by mapping each continuous value to the nearest discrete level. The resolution of a quantizer depends on the number of bits, B :

- **Quantization Levels (L):** The number of discrete amplitude steps available:

$$L = 2^B \quad (4)$$

- **Step Size (Δ):** The amplitude difference between adjacent levels for a full-scale range R (from $-V_{ref}$ to $+V_{ref}$):

$$\Delta = \frac{R}{2^B} \quad (5)$$

- **Quantization Noise:** The rounding error $e[n] = x_q[n] - x[n]$ is bounded by $-\Delta/2 \leq e[n] \leq \Delta/2$. Under uniform assumptions, the average noise power is:

$$\sigma_e^2 = \frac{\Delta^2}{12} \quad (6)$$

- **Signal-to-Quantization-Noise Ratio (SQNR):** For a full-scale sinusoidal input, the SQNR is:

$$\text{SQNR (dB)} \approx 6.02B + 1.76 \text{ dB} \quad (7)$$

This implies that each additional bit of resolution increases the SQNR by approximately 6 dB, representing a significant reduction in noise power and stored data fidelity.

3 Discrete-Time (DT) Signals & Classification

A discrete-time signal $x[n]$ is defined only at integer values of the independent variable n (sample index). Mathematically, it is a sequence of numbers:

$$x[n] = \{x[-1], \underline{x[0]}, x[1], x[2], \dots\} \quad (8)$$

3.1 Key Classifications

1. **Periodic vs. Aperiodic:** A signal $x[n]$ is periodic if there exists an integer $N > 0$ such that $x[n + N] = x[n]$ for all n . For a sinusoidal signal $A \cos(\omega_0 n)$, it is periodic only if:

$$\frac{\omega_0}{2\pi} = \frac{m}{N} \quad \text{where } m, N \in \mathbb{Z} \quad (9)$$

If the frequency ratio is irrational, the signal is aperiodic.

2. **Even vs. Odd Symmetry:**

- Even: $x[-n] = x[n]$ (Symmetrical about the y-axis).
- Odd: $x[-n] = -x[n]$ (Antisymmetrical; implies $x[0] = 0$).

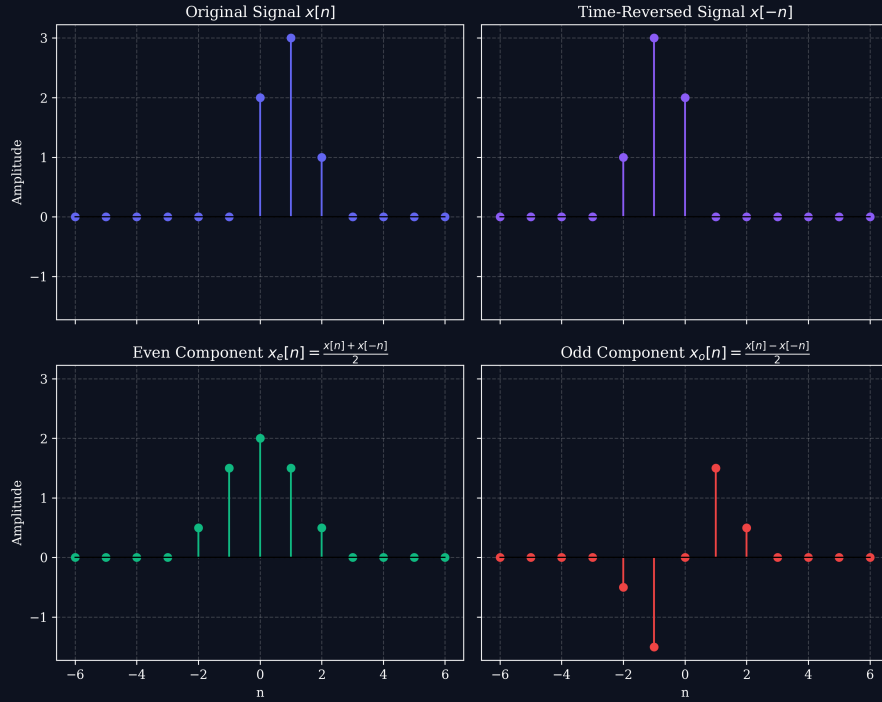


Figure 3: Decomposition of an asymmetric sequence $x[n]$ into its even component $x_e[n]$ and odd component $x_o[n]$.

3. **Energy vs. Power:**

- Total Energy: $E = \sum_{n=-\infty}^{\infty} |x[n]|^2$.
- Average Power: $P = \lim_{N \rightarrow \infty} \frac{1}{2N+1} \sum_{n=-N}^N |x[n]|^2$.
- Energy Signal: $0 < E < \infty \implies P = 0$.
- Power Signal: $0 < P < \infty \implies E = \infty$.

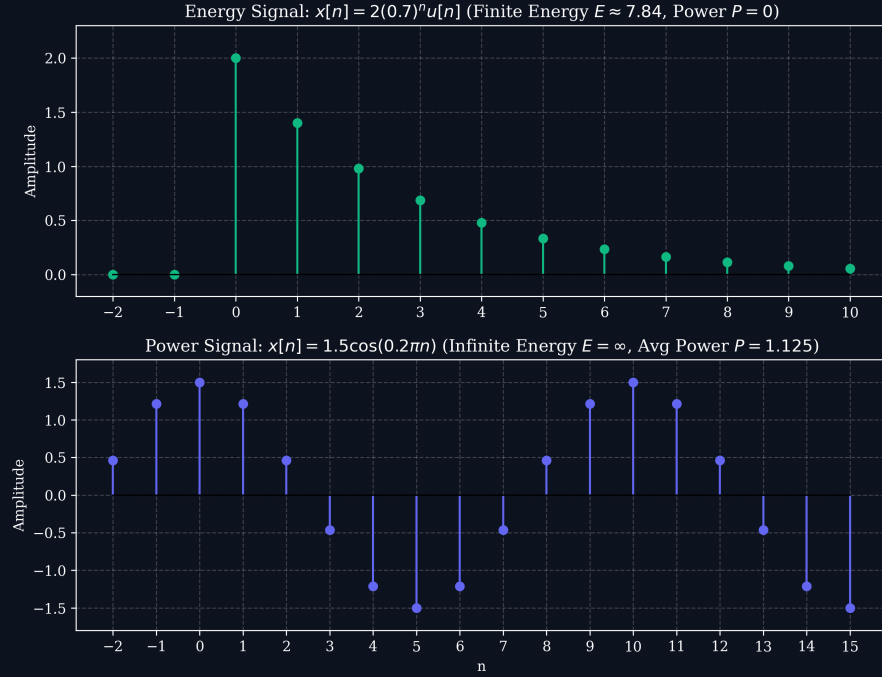


Figure 4: Visual comparison of an exponential decay (Energy Signal) and a periodic sinusoid (Power Signal).

4. Causality:

- Causal: $x[n] = 0$ for $n < 0$.
- Non-causal: $x[n] \neq 0$ for some $n < 0$.
- Anti-causal: $x[n] = 0$ for $n \geq 0$.

4 Elementary Sequences

These fundamental sequences act as building blocks to represent complex signals.

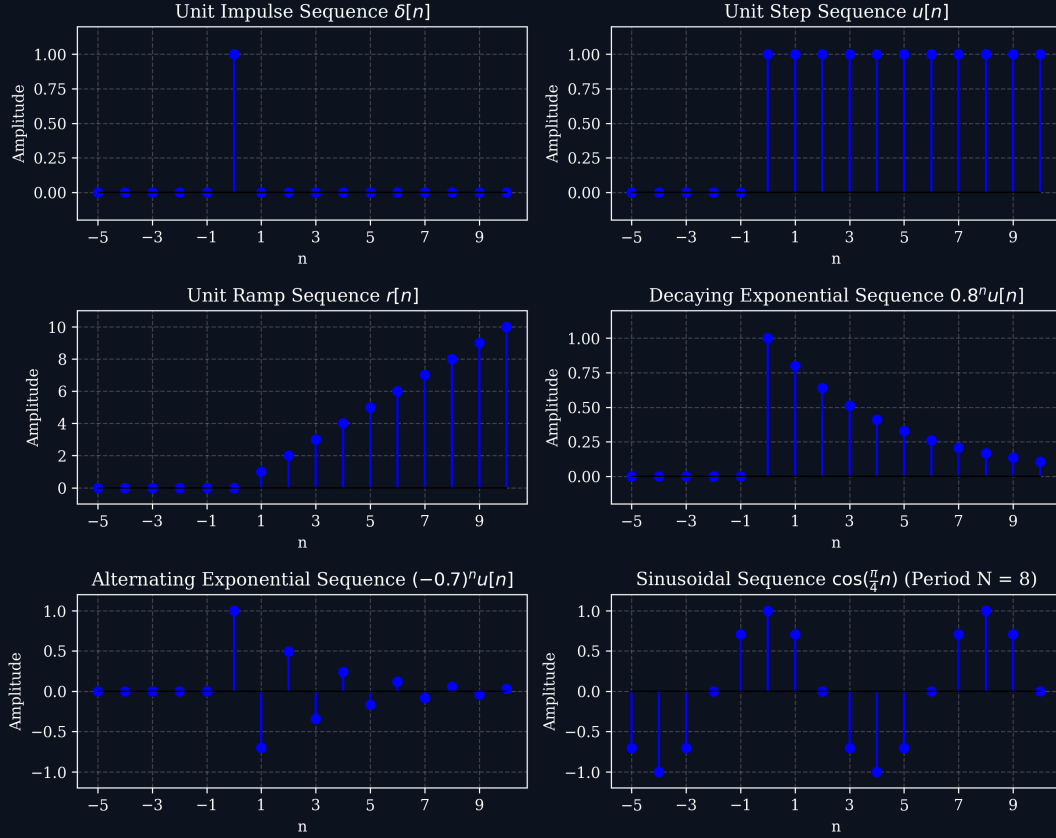


Figure 5: Graphical representations of elementary discrete-time sequences.

- **Unit Impulse $\delta[n]$:**

$$\delta[n] = \begin{cases} 1, & n = 0 \\ 0, & n \neq 0 \end{cases} \quad (10)$$

- **Unit Step $u[n]$:**

$$u[n] = \begin{cases} 1, & n \geq 0 \\ 0, & n < 0 \end{cases} \quad (11)$$

- **Unit Ramp $r[n]$:**

$$r[n] = n \cdot u[n] = \begin{cases} n, & n \geq 0 \\ 0, & n < 0 \end{cases} \quad (12)$$

- **Exponential Sequence $x[n] = a^n u[n]$:** (Decaying if $|a| < 1$, growing if $|a| > 1$).
- **Sinusoidal Sequence $x[n] = A \cos(\omega_0 n + \theta)$:** Where ω_0 is the digital frequency in radians/sample.

4.1 Operations and Properties of Signals

Discrete-time sequences can be manipulated using basic mathematical operations:

- **Time Shifting:** Shifting the signal index by an integer n_0 :

$$y[n] = x[n - n_0] \quad (13)$$

If $n_0 > 0$, the signal is delayed (shifted right). If $n_0 < 0$, it is advanced (shifted left).

- **Time Reversal (Folding):** Reflecting the signal about $n = 0$:

$$y[n] = x[-n] \quad (14)$$

- **Time Scaling (Decimation & Interpolation):**

- Decimation (Downsampling): $y[n] = x[D \cdot n]$, where $D \in \mathbb{Z}^+$ keeps every D -th sample.
- Interpolation (Upsampling): $y[n] = x[n/I]$ if n is a multiple of I , and 0 otherwise.

- **Arithmetic Operations:** Summing ($y[n] = x_1[n] + x_2[n]$), product ($y[n] = x_1[n] \cdot x_2[n]$), or amplitude scaling ($y[n] = A \cdot x[n]$).

4.2 Definition of a DSP System

A **system** is defined mathematically as an operator or transformation $\mathcal{T}$ that maps an input sequence $x[n]$ to an output sequence $y[n]$:

$$y[n] = \mathcal{T}\{x[n]\} \quad (15)$$

The digital signal processor is the physical implementation of this system. Systems are classified by these fundamental properties:

- **Linearity:** Satisfies both **Additivity (Superposition)** and **Homogeneity (Scaling)**.

1. Superposition (Additivity):

$$\mathcal{T}\{x_1[n] + x_2[n]\} = \mathcal{T}\{x_1[n]\} + \mathcal{T}\{x_2[n]\} \quad (16)$$

$$\begin{array}{ccccc} x_1[n] & \rightarrow & \boxed{\mathcal{T}} & \rightarrow & y_1[n] \\ x_2[n] & \rightarrow & \boxed{\mathcal{T}} & \rightarrow & y_2[n] \end{array}$$

$$\text{Outputs are added: } y_A[n] = y_1[n] + y_2[n]$$

$$\hline x_1[n] + x_2[n] \rightarrow \boxed{\mathcal{T}} \rightarrow y_B[n] = \mathcal{T}\{x_1[n] + x_2[n]\}$$

$$\text{Linear system condition: } y_A[n] = y_B[n]$$

2. Homogeneity (Scaling):

$$\mathcal{T}\{a \cdot x[n]\} = a \cdot \mathcal{T}\{x[n]\} \quad (17)$$

$$\begin{array}{ccccc} x[n] & \rightarrow & \boxed{\text{Scale by } a} & \rightarrow & \boxed{\mathcal{T}} \rightarrow y_A[n] \\ x[n] & \rightarrow & \boxed{\mathcal{T}} & \rightarrow & \boxed{\text{Scale by } a} \rightarrow y_B[n] \end{array}$$

$$\text{Linear system condition: } y_A[n] = y_B[n]$$

Mathematical Proof: Why a Straight Line Equation is Non-Linear

Consider the system:

$$y[n] = \mathcal{T}\{x[n]\} = m \cdot x[n] + c \quad (c \neq 0) \quad (18)$$

1. Homogeneity Check: For input $x_1[n] = a \cdot x[n]$:

$$y_1[n] = \mathcal{T}\{a \cdot x[n]\} = m(a \cdot x[n]) + c = amx[n] + c \quad (19)$$

Scaling the output by a :

$$a \cdot y[n] = a(m \cdot x[n] + c) = amx[n] + ac \quad (20)$$

Since $c \neq ac$ for $c \neq 0$ (when $a \neq 1$), homogeneity is violated.

2. **Additivity Check:** For input $x_3[n] = x_1[n] + x_2[n]$:

$$y_3[n] = m(x_1[n] + x_2[n]) + c = mx_1[n] + mx_2[n] + c \quad (21)$$

Sum of individual outputs:

$$y_1[n] + y_2[n] = (mx_1[n] + c) + (mx_2[n] + c) = mx_1[n] + mx_2[n] + 2c \quad (22)$$

Since $c \neq 2c$ for $c \neq 0$, additivity is violated.

Thus, the straight-line affine relation $y[n] = mx[n] + c$ is a **non-linear** system.

- **Time-Invariance:** A time-shift of the input causes an identical time-shift of the output:

$$\mathcal{T}\{x[n - n_0]\} = y[n - n_0] \quad (23)$$

$$\begin{array}{ccccc} x[n] & \rightarrow & \boxed{\text{Delay } n_0} & \rightarrow & \boxed{\mathcal{T}} \rightarrow y_A[n] \\ x[n] & \rightarrow & \boxed{\mathcal{T}} & \rightarrow & \boxed{\text{Delay } n_0} \rightarrow y_B[n] \end{array}$$

Time-Invariance condition: $y_A[n] = y_B[n]$

- **Causality:** The output at index n_0 depends only on present and past input values ($x[n]$ for $n \leq n_0$).
- **Stability:** A system is BIBO (Bounded-Input Bounded-Output) stable if every bounded input ($|x[n]| \leq M_x < \infty$) produces a bounded output ($|y[n]| \leq M_y < \infty$).

5 Checkpoint & Class Discussion Questions

1. **Q1:** Determine if $x[n] = \cos(\frac{3}{10}n)$ is periodic or aperiodic.

- **Solution:** $\omega_0 = 3/10$. Thus, $\frac{\omega_0}{2\pi} = \frac{3}{20\pi}$. Since π is irrational, this ratio is irrational, meaning the signal is **aperiodic**.

2. **Q2:** Write the mathematical relationship between the unit step $u[n]$ and unit impulse $\delta[n]$.

- **Solution:** $\delta[n] = u[n] - u[n - 1]$ and $u[n] = \sum_{k=0}^{\infty} \delta[n - k]$.